Batch: A2	Roll No.: 16010324044
Experiment / assignment / tutorial No. 11	
Grade: AA / AB / BB / BC / CC / CD /DD	
Signature of the Staff In-charge with date	

TITLE: Introduction to packet Analyser tool -Wireshark

AIM: To i) Capture the live network traffic
ii) observe and study the PDUs of different TCP/IP layers

OUTCOME: Student will be able to understand layered architecture of communication network and physical layer services and study the PDUs of various layers.

Theory:

Wireshark is a open source software tool used to monitor the network traffic through a network interface. It is the most widely used network monitoring tool today. Wireshark is loved equally by system administrators, network engineers, network enthusiasts, network security professionals and black hat hackers.

The extent of its popularity is such, that experience with Wireshark is considered as a valuable/essential trait in a computer networking-related professional.

Features of Wireshark Tool:

- Deep inspection of hundreds of protocols, with more being added all the time
- Live capture and offline analysis
- Standard three-pane packet browser
- Multi-platform: Runs on Windows, Linux, macOS, FreeBSD, NetBSD, and many others
- Captured network data can be browsed via a GUI, or via the TTY-mode TShark utility
- The most powerful display filters in the industry
- Rich VoIP analysis
- Read/write many different capture file formats: tcpdump (libpcap), Pcap NG, Catapult DCT2000, Cisco Secure IDS iplog, Microsoft Network Monitor, Network General Sniffer® (compressed and uncompressed), Sniffer® Pro, and NetXray®, Network Instruments Observer, NetScreen snoop, Novell LANalyzer, RADCOM WAN/LAN Analyzer, Shomiti/Finisar Surveyor, Tektronix K12xx, Visual Networks Visual UpTime, WildPackets EtherPeek/TokenPeek/AiroPeek, and many others
- Capture files compressed with gzip can be decompressed on the fly
- Live data can be read from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI, and others (depending on your platform)
- Decryption support for many protocols, including IPsec, ISAKMP, Kerberos, SNMPv3, SSL/TLS, WEP, and WPA/WPA2
- Coloring rules can be applied to the packet list for quick, intuitive analysis Output can be exported to XML, PostScript®, CSV, or plain text.

The basic features of Wireshark are:

- **Packet Monitor:** This segment visually shows the packets flowing inside the network. There are color codes for each type of packet. The packets are shown with the following information :
  1. Source address
  2. Destination address
  3. Packet type
  4. Hex dump of the packet
  5. Contents of the packet in text
  6. Source port(if applicable)
  7. Destination port(if applicable)
- **Import from a capture file:** This feature lets you import packets dump from a capture file to analyse further.

Procedure:

1. Launch Wireshark, select an interface(select one that is currently communicating, which can be verified by the zigzag pattern in front of the name of the interface)
2. Click on the fin icon to start capturing packets.
3. Select any of the packets like TCP, UDP etc from the live traffic.
4. Study the details of various layers PDUs such as for Transport Layer-TCP or/UDP Headers, for Network layer-IP Header, for Data Link Layer-Ethernet Frame .
5. Extract the information for each PDU like for TCP/UDP -Source and Destination Ports, sequence number and other fields.
6. For IP packet: Source and Destination IP addresses , and all other fields of IP haeder.
7. For Ethernet frame Sourse ,destination MAC addresses and other fields.
8. Attach all the relevant screenshots including all the PDUs.

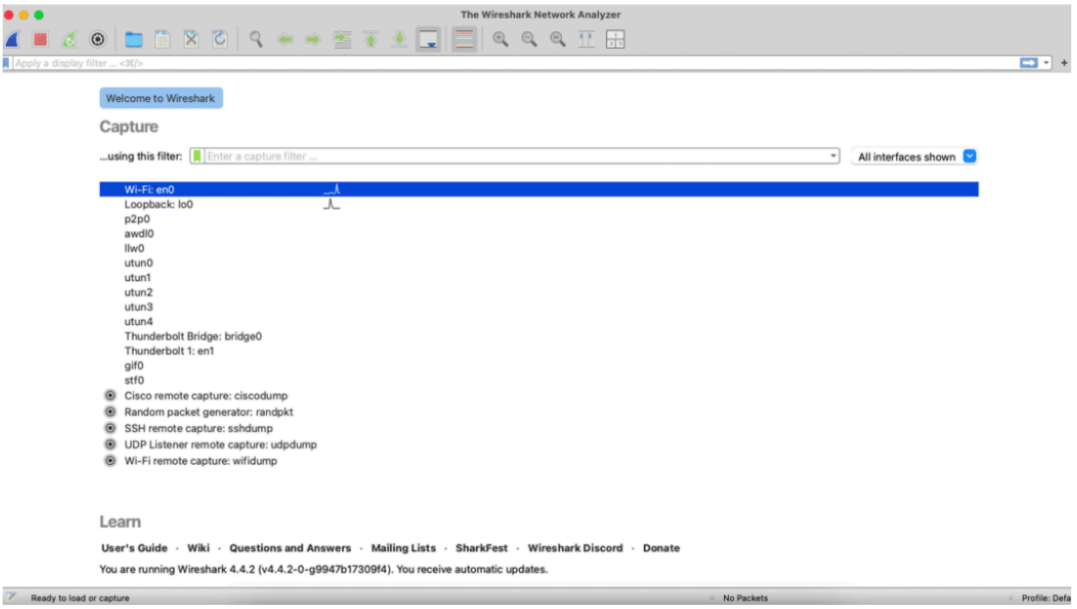


Fig 1: Wireshark UI

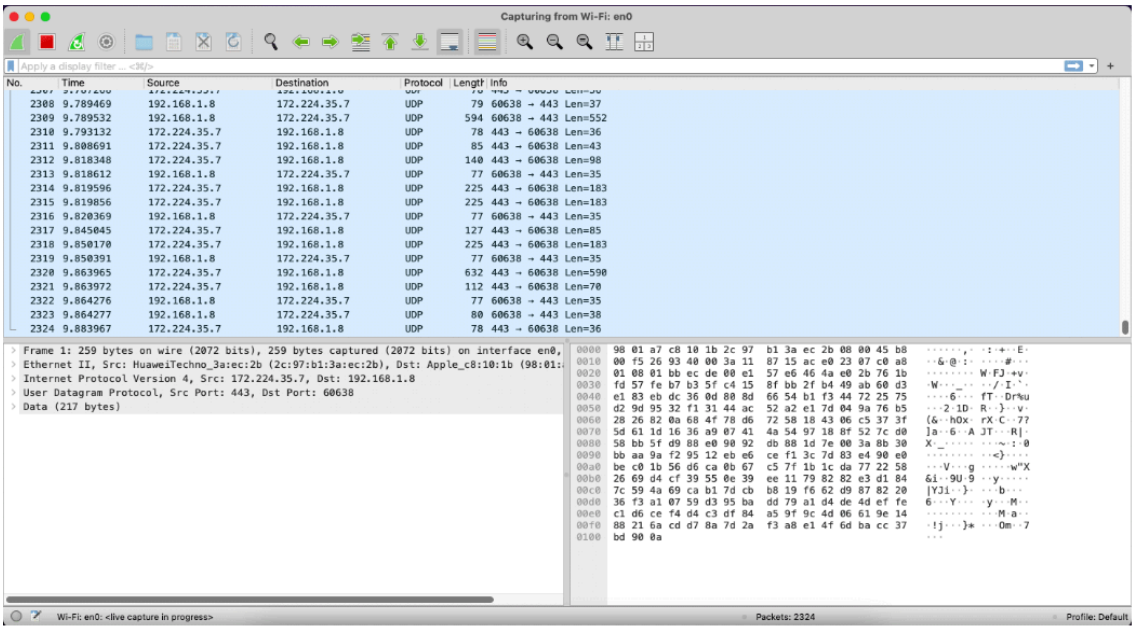
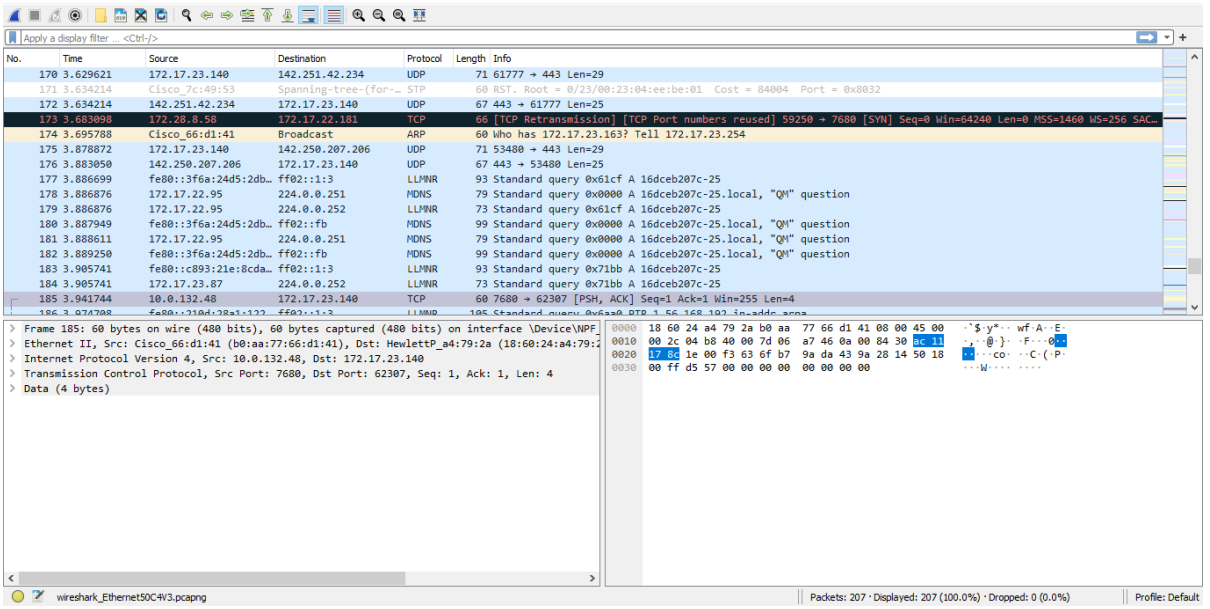


Fig 2: Packets captured on wireshark

Observations:

Layer	PDU	Important fields observed
Data Link Layer	Ethernet Frame	Source MAC, Destination MAC, Type
Network Layer	IP Packet	Source IP, Destination IP, TTL, Protocol
Transport Layer	TCP Segment	Source Port, Destination Port, Sequence Number, Flags
Application Layer	Application Data	Protocol/application information

Screenshots:




```
Frame 185: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\N
    Section number: 1
  > Interface id: 0 (\Device\NPF_{B4C359E8-8EA3-48FC-B859-CB5B5CF3E91C})
    Encapsulation type: Ethernet (1)
    Arrival Time: Sep 24, 2026 01:55:57.010426000 Pacific Daylight Time
    [Time shift for this packet: 0.000000000 seconds]
    Epoch Time: 1790240157.010426000 seconds
    [Time delta from previous captured frame: 0.036003000 seconds]
    [Time delta from previous displayed frame: 0.036003000 seconds]
    [Time since reference or first frame: 3.941744000 seconds]
    Frame Number: 185
    Frame Length: 60 bytes (480 bits)
    Capture Length: 60 bytes (480 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:data]
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
```

```
Ethernet II, Src: Cisco_66:d1:41 (b0:aa:77:66:d1:41), Dst: HewlettP_a4:79:2a (18:60:24:a4:7
  > Destination: HewlettP_a4:79:2a (18:60:24:a4:79:2a)
  > Source: Cisco_66:d1:41 (b0:aa:77:66:d1:41)
    Type: IPv4 (0x0800)
    Padding: 0000
```

```
Internet Protocol Version 4, Src: 10.0.132.48, Dst: 172.17.23.140
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 44
    Identification: 0x04b8 (1208)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 125
    Protocol: TCP (6)
    Header Checksum: 0xa746 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.0.132.48
    Destination Address: 172.17.23.140
```

```
Transmission Control Protocol, Src Port: 7680, Dst Port: 62307, Seq: 1, Ack: 1, Len: 4
    Source Port: 7680
    Destination Port: 62307
    [Stream index: 9]
    [Conversation completeness: Incomplete (12)]
    [TCP Segment Len: 4]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 1874303706
    [Next Sequence Number: 5    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1134176276
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
    Window: 255
    [Calculated window size: 255]
    [Window size scaling factor: -1 (unknown)]
    Checksum: 0xd557 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
    TCP payload (4 bytes)
```

```
Data (4 bytes)
    Data: 00000000
    [Length: 4]
```

Post Lab Questions:

1. Draw IP header format ,TCP Header format, UDP header format and Ethernet frame format.

CONCLUSION

In this experiment, I successfully captured and analyzed live network traffic using Wireshark. The captured packets were studied at different TCP/IP layers, including the Ethernet frame at the Data Link Layer, IP packet at the Network Layer, and TCP/UDP segment at the Transport Layer. Important fields such as source and destination MAC addresses, IP addresses, port numbers, sequence numbers and other header information were observed. Thus, the experiment helped in understanding the layered architecture of computer networks and the structure of PDUs at different TCP/IP layers.

Signature of faculty in-charge